A Neutrino Corridor for a 10 TeV Muon Collider at Fermilab

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The decay neutrinos of a multi-TeV muon collider form pencil beams of unprecedented intensity and energy, and their interaction with the surrounding landscape — where they graze the Earth's surface — is a recognized siting constraint. We invert the constraint into an asset. We show that an 11 km collider ring fits the Fermilab campus with its interaction-point straight on the true-north meridian $-88.222972°$, tilted upward by 15.4 mrad (one LEP of ring-plane inclination). The north-going neutrino beam then leaves the ground *inside the laboratory fence*, delivering $\sim 10^{12}$ TeV-scale neutrino interactions per year to a one-tonne on-site experiment before climbing a utility right-of-way into federally navigable airspace; the south-going beam dives beneath the region's aquifers to a perigee of 789 m and resurfaces 198 km away on University of Illinois agricultural land, where its exit strip doubles as a second neutrino experiment of $\sim 3 \times 10^8$ interactions per tonne-year. The second interaction point's south beam exits onto the same university's airport. Every point at which a beam passes within 500 ft of the ground surface lies on federal, utility-easement, or university land, and every potable-aquifer crossing beyond the laboratory fence is bounded at a factor $\gtrsim 45$ — typically orders of magnitude — below drinking-water limits. The alignment costs $\sim 11\%$ of one accelerator ring's circumference and one LEP-scale plane tilt, and converts the muon collider's defining radiological liability into two shared-science facilities.

Introduction.—A muon collider at $\sqrt{s} = 10$ TeV is the most compact route to the energy frontier now under serious study [1, 2]. Its defining environmental feature is also its most distinctive experimental opportunity: every muon decays, and the decays in each long straight section emit neutrinos in a cone of half-angle $1/\gamma \simeq 21$ μrad (at $E_\mu = 5$ TeV) along the straight's axis. Where that pencil intersects the Earth's surface tens of kilometres away, the showers from neutrino interactions in the final metres of soil can produce annual doses approaching regulatory limits — the "neutrino radiation" problem first quantified by King [3] and adopted as a driving constraint by the International Muon Collider Collaboration, whose reference siting aims its insertion plumes at the Mediterranean and a fenced mountainside [2]. The same pencil, sampled *before* it exits, is the most intense high-energy neutrino beam ever conceived, with fully known flavor composition and parent kinematics [4].

This Letter reports a complete siting geometry for the Fermilab campus in which both properties are engineered simultaneously and deliberately: the beam's two ground intersections are *chosen*, both fall on institutional land, both host experiments, and the intervening 200 km of private land is crossed only at altitudes or depths at which no surface owner's use of their property is touched.

The geodetic coincidence.—Three surveyed facts make the geometry work. (i) A commercial transmission right-of-way (ComEd *Aurora–Wayne*) runs almost exactly due north–south at longitude $\simeq -88.223°$ from the Fermilab boundary northward. (ii) The northeast corner of the laboratory's site polygon lies on that meridian to within one metre of longitude. (iii) Along the meridian $-88.222972°$ the site offers 5.48 km of north–south extent — room for an 11 km racetrack's 700 m straights and 1.5 km-radius arcs, and for the acceleration chain's straights on the same line. Because a due-north geodesic holds constant longitude, aligning every long straight with the corridor reduces to placing it on this meridian at azimuth 0°.

Machine geometry.—We fit an 11.0 km racetrack collider ($R_{\rm arc} = 1528$ m, 2×700 m insertions) against the real site boundary with a 150 m setback, its east straight on the corridor meridian and its interaction point (IP) at 41.84428° N, placed such that both ends of the straight and both beam emergences remain well inside the boundary. The ring plane is tilted $\theta_0 = 15.40$ mrad (0.88°) upward toward north — 10% beyond LEP's built precedent of 14 mrad — with the straightaway at 191 m above sea level, 34 m below local grade, at the glacial-drift/Silurian-dolomite contact. The rapid-cycling-synchrotron chain (6.28 and 14.72 km racetracks at 60–80 m depth) shares the meridian with mrad-scale plane pitches so that all straight plumes converge on one on-site detector volume; the alignment's only machine cost is a 10.8% circumference shortfall in the largest RCS relative to an unconstrained boundary-hugging fit.

In a spherical-geoid frame a straight beam launched at height h_0 with elevation angle θ_0 has height $h(s) = h_0 + \theta_0 s + s^2/2R_\oplus$ above the curving surface. Northward, with real (SRTM) terrain, the beam surfaces at 41.8640° N — 2.2 km from the IP and 655 m *inside* the laboratory fence — crosses the boundary 8 m above grade, climbs the right-of-way, passes the nearest residential area 116 m (380 ft) above grade — 350 ft above a 10 m roofline — and enters federally navigable airspace (500 ft above ground) 9.3 km beyond the fence. Southward the same straight aims 15.40 mrad below the horizon: the beam passes beneath the Aurora conurbation at 74–152 m depth, reaches a perigee of 789 m in the brackish upper Mt. Simon sandstone — the horizon northern Illinois uses for natural-gas storage, sealed beneath the Eau Claire aquitard and below every potable unit — and re-emerges $2\theta_0$ of central angle later: 198 km south, at 40.067° N, on the University of Illinois Urbana–Champaign (UIUC) *South Farms*, approaching under the city of Urbana at 66–119 m depth. A ±0.1 mrad trim moves this exit ±1.3 km along the meridian: targeting a chosen parcel is a survey problem, not a physics one. The diametrically opposite second IP, on the ring's west straight (meridian −88.2598°), inherits the same plane tilt; its south beam exits at ≃40.045° N on Willard Airport — owned and operated by the same university — after a +0.13 mrad trim well inside the ±1 mrad mover budget already carried by the IMCC design [2].

The civic envelope.—The siting rule that organizes the study is geometric: wherever either beam is within 500 ft (152 m) of the ground surface, vertically, the surface land must belong to an institutional partner — the Department of Energy, the utility easement, or the university. Above 500 ft over uncongested land an aircraft may legally fly (14 CFR 91.119); since *United States v. Causby* (1946) a landowner's airspace claim reaches only the "immediate reaches" they can use, so a silent, invisible, depositing-nothing neutrino pencil hundreds of feet up is comfortably in the public highway. Below 500 ft of rock the beam is deeper than every private well (the shallow Silurian aquifer's wells run 30–90 m), and — unlike the deep tunnels bored under Chicago through thousands of recorded subsurface easements — nothing is built: neutrinos simply pass. Auditing the baseline against this rule leaves four honest exceptions, all deep or high crossings, none touching the surface: 1.9 km of overflight at 384–500 ft over low-density countryside; 5.6 km at 74–152 m under Aurora's east edge; and the farmland-plus-Urbana approach at 66–152 m. Standard avigation or subsurface easements close each, though none plausibly requires it.

Fluxes and rates.—With IMCC-class beam parameters (1.8×10^{12} muons per bunch per sign at 5 Hz, 1.2×10^{7} s operational year, 90% transmission), 6.2×10^{18} decays per year are aimed along each direction of the IP straight (700 m of an 11 km ring). The on-axis flux at distance L is $\Phi = N\gamma^2/\pi L^2$; with $\sigma_\nu \simeq 0.35\times10^{-38}\,\mathrm{cm}^2\,\mathrm{GeV}^{-1}\times E_\nu$ per nucleon and $\langle E_\nu\rangle \simeq 0.65\,E_\mu$ on axis, an on-site detector 1.3 km from the IP intercepts $\sim7\times10^{12}$ charged- and neutral-current interactions per tonne-year on axis (CSMS cross-sections, both species per decay) at $\langle E_\nu\rangle \simeq$ 3.2 TeV — eight orders of magnitude beyond present TeV-neutrino samples, in a beam whose flavor content ($\nu_\mu\bar{\nu}_e$ or $\bar{\nu}_\mu\nu_e$, selectable by circulation direction) is exactly known. At the South Farms exit, 198 km out, the surviving flux still delivers $\sim3\times10^{8}$ interactions per tonne-year: the exit strip that must be controlled anyway becomes a far detector site with L/E two orders of magnitude beyond any accelerator baseline, on land owned by a research university.

Beam size.—Because muon decay is isotropic in the muon rest frame, the lab-frame profile is the beamed form $dN/d\Omega = N\gamma^2/[\pi(1+\gamma^2\theta^2)^2]$, whose on-axis value reproduces the flux above. The profile is therefore self-similar, with the single scale L/γ: exactly half the neutrinos fall inside $1/\gamma$, 90% inside $3/\gamma$. The boost also correlates energy with angle, $\langle E\rangle \propto (1+\gamma^2\theta^2)^{-1}$, so with $\sigma_\nu \propto E$ the interaction density falls as $(1+\gamma^2\theta^2)^{-3}$ and is appreciably tighter: 75% of interactions inside $1/\gamma$ and 99% inside $3/\gamma$. The muon beam's own divergence, $\sigma_\theta(s) = \sqrt{\epsilon_N/\gamma\beta(s)}$ with $\beta(s) = \beta^* + s^2/\beta^*$, is 0.59 mrad at the IP for the reference optics ($\epsilon_N = 25\ \mu$m, $\beta^* = 1.5$ mm) but falls below $1/\gamma$ within ±4 cm of the collision point — 10^{-4} of the insertion — leaving a halo four orders of magnitude below the core in surface density. *The far-field spot is purely kinematic*, and varying β^* over two orders of magnitude does not move it. Numerically the 50% radius is 2.7 cm at the on-site detector, 6 cm at the fence, and 4.2 m at the southern exit. Two consequences follow. A tungsten cylinder 16 cm in diameter and 2.4 m long — one tonne — intercepts 99% of the interactions, so the headline rate describes a portable object rather than a cavern. And at the exits the circular cross-section is stretched along the ground by $1/\sin(\text{graze})$: the northern emergence paints a 9 cm × 5.9 m streak inside the laboratory fence, while the southern one is an 8.3 m × 534 m strip, which is the actual geometry of the land-use question.

Radiological audit.—Dose: everywhere along the corridor except the two chosen exits the beam is deep underground or high overhead, and the exit strips themselves are the entire land-use footprint; at the southern exit the $1/L^2$ dilution over 198 km brings even King's deliberately conservative equilibrium model [3] to ∼0.5 mSv/yr mitigated in a strip a few metres wide on fenced farmland (the framework, corrections, and the deep-reference alternative are in the accompanying study [5]). Groundwater: neutrinos activate nothing directly; the only pathway is the hadronic showers of the $\sim2\times10^{15}$ interactions per year distributed along the full 198 km chord (3.5×10^{-4} interaction lengths of rock), yielding $\sim7\times10^{16}$ tritium atoms per year (≃0.06 Ci/yr at saturation) with a $1/L^2$ weighting toward the IP. In the maximally conservative limit — groundwater stagnant inside the centimetres-wide beam core at each aquifer's closest approach, for many tritium lifetimes, with zero exchange — the EPA drinking-water limit (740 Bq/L) is exceeded only within 2.1 km of the IP, entirely inside the laboratory boundary. Traced through a dipping stratigraphic model of the Illinois Basin's flank (unit tops anchored to NuMI construction geology, ISGS mapping, and the region's gas-storage fields), the municipal sandstone aquifers are crossed ∼14 km and ∼28 km out and ceiling at 45× and 190× below the limit; the chord bottoms in the brackish Mt. Simon; and the drift sands near Champaign–Urbana ceiling at $\sim10^4\times$ below. Groundwater advection — metres per day in the karstic on-site dolomite — lowers all of these by further orders of magnitude.

Conclusion.—The neutrino radiation problem of a 10 TeV muon collider at Fermilab can be solved by construction rather than mitigation: one LEP of ring-plane tilt and a meridian alignment place both surface intersections of the collision straight — and both of the second interaction point's — on federal or university land, thread the 200 km between them outside the envelope of private use, bound aquifer activation orders of magnitude below drinking-water standards, and produce two neutrino experiments of unprecedented reach as a by-product. The geometry uses only surveyed data and analytic conservative models; a real lattice, FLUKA/MARS transport, and hydrogeological modelling are the obvious next steps, and all inputs, code, and an interactive reproduction of every number in this Letter are public [5].

Geometry, terrain, and boundary data: OpenStreetMap contributors (ODbL) and NASA SRTM. This concept study used no beam time and harmed no groundwater.

[1] K. M. Black *et al.*, "Muon Collider Forum report," JINST **19**, T02015 (2024), arXiv:2209.01318.

[2] C. Accettura *et al.* (International Muon Collider Collaboration), "Interim report for the International Muon Collider Collaboration," CERN Yellow Rep. Monogr. (2024), arXiv:2407.12450.

[3] B. J. King, "Potential hazards from neutrino radiation at muon colliders," arXiv:physics/9908017.

[4] L. Bojorquez-Lopez, M. Hostert, T. Menzo, and J. Zupan, "The neutrino slice at muon colliders," arXiv:2412.14115.

[5] L. Lee, "FNAL muon collider corridor siting study," https://lawrenceleejr.github.io/FNALMuonColliderSite/ and https://github.com/lawrenceleejr/FNALMuonColliderSite (2026).